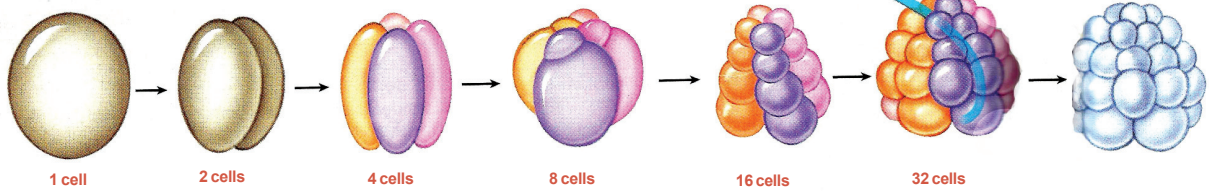
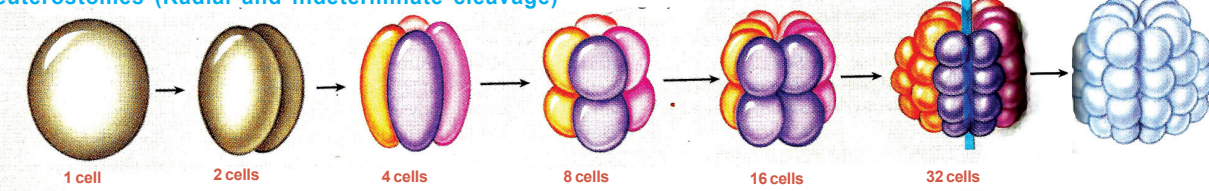


Diagrams showing embryonic development in protostomes and deuterostomes.

Protostomes (Spiral and determinate cleavage)



Deuterostomes (Radial and indeterminate cleavage)



1.10 BIODIVERSITY

I. What is Biodiversity?

When we observe our surroundings we find different kinds of organisms which vary in size, form, feeding habits, behaviour, etc. For example there are more than 20,000 species of ants, 3,00,000 species of beetles, 28,000 species of fishes and 20,000 species of orchids. This variation of life at various levels of biological organization is termed as biodiversity.



II. Levels Of Biodiversity

Biodiversity exists not only at the species level but at all levels of biological organization ranging from macromolecules within the cells to **biomes** (biotic community in a large area).

The term biodiversity was popularized by the sociobiologist **Edward Wilson** to describe the combined diversity at all levels of biological organization. The three levels of biodiversity are:

1. Genetic diversity
2. Species diversity
3. Ecological diversity.

1. Genetic diversity

It is the diversity of genes within a species. A single species may show high diversity at the genetic level over its distributional

range. For e.g. *Rauwolfia vomitoria*, a medicinal plant growing in the Himalayan ranges shows great genetic variation, which might be in terms of **potency** and **concentration** of the active chemical (**reserpine** extracted from it is used in treating high blood pressure) that the plant produces. India has more than 50,000 different strains of rice, and 1,000 varieties of mangoes. **Genetic diversity** increases with environmental variability and is advantageous for its survival.

2. Species Diversity

It is the diversity at the species level. e.g: amphibian diversity in the Western Ghats is greater than that of the Eastern Ghats.

3. Ecological diversity

Diversity at a higher level of organization, i.e. at the ecosystem level is called 'Ecological diversity'. e.g: India with its deserts, rain forests, mangroves, coral reefs, wet lands, estuaries and alpine meadows has greater ecosystem diversity than many other countries such as the Scandinavian country Norway.

The three indices of ecological diversity are- **Alpha**, **Beta** and **Gamma** diversity

i. Alpha diversity: It is measured by counting the number of taxa (usually species) within a particular area, community or ecosystem.

ii. Beta diversity: It is the species diversity between two adjacent ecosystems and is obtained by comparing the number of taxa **unique to each of the ecosystems**.

iii. Gamma diversity: It is the measure of the **overall diversity for different ecosystems** within an ecological region ('**Ecological Region**' is a large area constituting a natural ecological community with characteristic flora and fauna, bounded by natural boundaries).

III. Other attributes of biodiversity

A species unique to a given area is called **endemic species**. Pattern of biodiversity depends on factors such as i) Latitude and ii) Species-area relationship

1. Latitudinal Gradient in Diversity

Biodiversity is not uniform throughout the world but shows rather uneven distribution. The most important pattern of biodiversity is latitudinal gradient in diversity. This means that there is an increasing diversity from the poles to the equator (terrestrial biodiversity increases from the poles to the equator). There is a vast majority of species concentrated in the tropics and sub tropical regions. This means localities at lower latitudes have more species than localities at higher latitudes.

Tropics harbour more species than temperate or polar areas for e.g. The tropical Amazon rain forest in south America has the greatest biodiversity on the Earth. Species diversity is more in the tropics. The following data explains latitudinal gradient :

Place	Number of species of birds	Latitude
Colombia	1400	0°N
New York	105	41°N
Green land	56	71°N

From the above data it is clearly evident that as the latitude increases the species diversity decreases.

Reasons for greater biodiversity in the tropics:

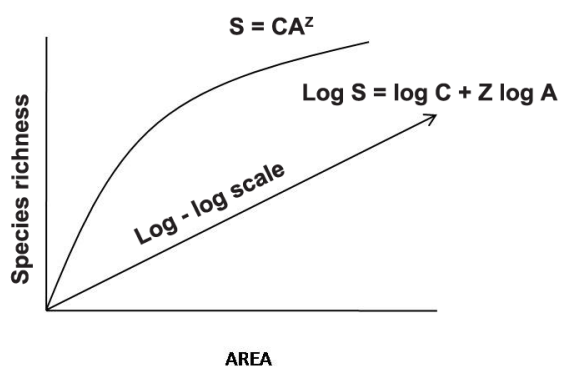
Reason 1: Tropical latitudes have remained relatively undisturbed for millions of years and thus had a long 'evolutionary time'. As long duration was available in this region for speciation, it led to the species diversification. (Note: The temperate regions were subjected to frequent glaciations in the past).

Reason 2: Tropical climates are relatively more constant and predictable than that of the temperate regions. Constant environment promotes **niche specialization** (how an organism responds, behaves with environment and other organisms of its biotic community), and this leads to greater species diversity.

Reason 3: Solar energy, resources like water etc., are available in abundance in this region. This contributed to higher productivity in terms of food production, leading to greater diversity.

2. Species-Area relationships

Before we learn relation between species richness and the area available to them, let us learn the term 'species richness'.



Species Richness is the number of species per unit area. The more the number of species in an area the more is the species richness.

Alexander von Humboldt observed that within a region, species richness increased with increasing explored area, but only up to a limit. In fact, the relation between species richness and area for a wide variety of taxa (angiosperm plants, birds, bats, fresh water fishes) turns out to be a non-linear curve. On a logarithmic scale, the relationship is a straight line described by the equation.

$$\log S = \log C + Z \log A$$

where

S = species richness

A = area

Z = slope of the line (regression coefficient)

C = Y-intercept

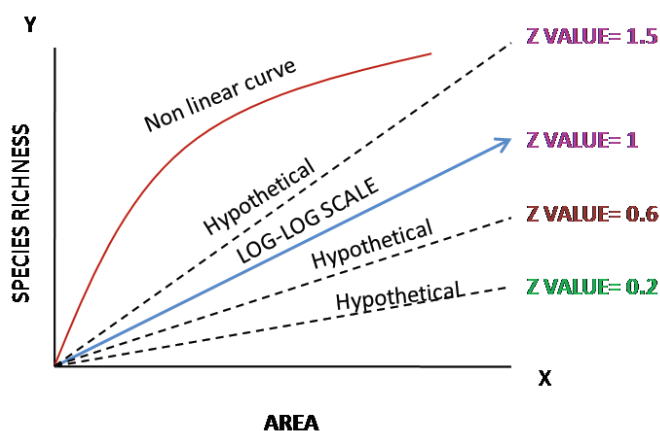
Ecologists have discovered that the value of Z lies in the range of 0.1 to 0.2 regardless of the taxonomic group or the region (whether it is California or New York or Britain). If you analyse species-area relationship among very large areas like entire continents the slope of the line is much steeper (Z values in the range of 0.6 to 1.2) for example for frugivorous (fruit-eating) birds and mammals in the tropical forests of the different continents, the slope is found to be **1.15**.

What do **steeper slopes** mean in this context ?

A simplified explanation to help Biology Teachers

Teachers of Biology might find it difficult to understand and explain the mathematical expression represented in this part. The graphic representation is essentially intended to explain the relationship between 'species richness' and 'area'. In a specified area the relation between ' S ' (species richness) and ' A ' (area) is $S = CA^Z$ and it is represented by a **Non-linear curve**.

For easy understanding, to get a **linear relation** between S and A , '**log scale**' is taken. Logarithmic relation between A and S is given by $\log S = \log C + Z \log A$. The '**curve**' shows that species richness increases with increase in area to a certain extent and approaches an **equilibrium state/ stable state**. It does not indicate the '**specific rate**' at which species richness increases with reference to area. The '**SLOPE**' of the graph drawn between $\log A$ (taken along **X axis**) and $\log S$ (taken along **Y axis**) is Z . Increase in the 'slope' indicates **increase in Z** indicating 'increase in species richness', as shown in the graph. For example, if the log-log scale makes 45° with X axis, Z is **1**. If the slope is less, Z is less than '**1**' and if the slope is more than 45° the value of Z is more than **1**. Now it is easy to understand that the value of Z in a '**tropical rain forest**' is more than 1, as represented by the 'increased slope'.



Importance of Species Diversity to the Ecosystems

Communities with more types of species (more biodiversity) tend to be more stable than those with less number of types of species. Stable communities generally withstand disturbances (natural or man-made). **Tilman's** experiments with 'outdoor plots' showed that 'plots with more species showed less variations in biomass year-to-year'. He also showed that increased **diversity contributed to higher productivity**.

What if we lose a few species? Will it affect man's life? **Paul Ehrlich's** experiments 'the **RIVET POPPER**' hypothesis, taking an aeroplane as an ecosystem, explains how removal of one by one 'rivets' (species of an ecosystem) of various parts can slowly damage the plane(ecosystem)-shows how important a 'species' is in the overall functioning of an ecosystem. Removing a rivet from a seat or some other relatively minor important parts may not damage the plane, but removal of a rivet from a part supporting the wing can result in a crash. Likewise, removal of a '**critical species**' may affect the entire community and thus the entire ecosystem.

IV. Role of biodiversity

Biodiversity is beneficial to human beings as it plays an important role at various levels of development and to explain the role played, the different aspects are categorised into three types,

1. Narrowly utilitarian argument
2. Broadly utilitarian argument
3. Ethical argument

1) Narrowly utilitarian argument: Human beings derive countless economic benefits from nature as biodiversity is a **reservoir of resources**. Food (cereals, pulses & fruits), Firewood, Fibre, Construction material, Industrial products (tannins, lubricants, dyes, resins, perfumes, rubber, latex, cork, etc.) and products of medicinal importance (for example Anticancer drugs – **Vinblastin** from *Vinca rosea*, **Digitalin** from the 'fox glove' plant (*Digitalis purpurea*) to treat certain cardiac problems, etc.) are obtained from diverse living organisms which are economically important. Bio-prospecting nations endowed with rich biodiversity can expect to reap enormous benefits.

2) Broadly utilitarian argument: It explains that biodiversity plays a major role in many ecosystem services that nature provides, for which we cannot put a price tag (*we cannot fix its price*). To exemplify this, consider the following -

e.g.1 : Amazon produces 20 % of the total oxygen in the earth's atmosphere.

e.g.2 : Pollination by pollinating agents(bees, birds, bats, etc.) without which plants do not produce fruits and seeds. There are other benefits also, the ones which cannot be measured in terms of money, like aesthetic pleasure derived by taking a walk in the woods.

- 3) **Ethical argument** : It relates to what we owe to plants, animals and microbe species with which we co-exist on this planet. The moral duty to care and pass on biological legacy to future generations is the need of the hour as every species has an *intrinsic value*.

V. Threats to Biodiversity

The following are the '**four major causes**' (**THE EVIL QUARTET**) for accelerated rates of species extinction in the world.

1. **Habitat Loss And Fragmentation** : These are the most important reasons for the loss of biodiversity.
 - a) Deforestation-leads to species extinction in forests e.g: tropical rain forests once covering 14% of the earth's land surface is now not more than 4%.
 - b) Conversion of forest land to agricultural land e.g: the AMAZON RAIN FOREST, called '**lungs of our planet**', harbouring innumerable species is cut and cleared to cultivate SOYA BEANS or conversion to grass lands for raising beef cattle.
 - c) Pollution enhances degradation of habitats and threatens the survival of many species as pollutants change the quality of the environment.
 - d) Fragmentation of habitat leads to population decline e.g: mammals and birds requiring large territories and certain animals with migratory habits are badly affected.

Fragmentation

It is the process of formation of discontinuities in the natural habitats due to geological processes or human activities. Geological fragmentation may lead to speciation, but fragmentation caused by man (human activity) mostly leads to extinction of many species.

2. **Over-exploitation: When need turns to greed it leads to over exploitation** e.g., Steller's sea cow (sea cow named after Steller, a naturalist), passenger pigeon (which existed in North America) are extinct due to over-exploitation by humans. The existence of many commercially important marine fishes are endangered as they are over harvested.
3. **Invasion Of Alien Species** : When alien species are introduced into a habitat, they turn **invasive** and establish themselves at the cost of the indigenous species (organisms which occur naturally in a particular region).

E.g.1: **Nile perch** introduced into Lake Victoria, in east Africa led to the extinction of 200 species of cichlid fish in the lake.

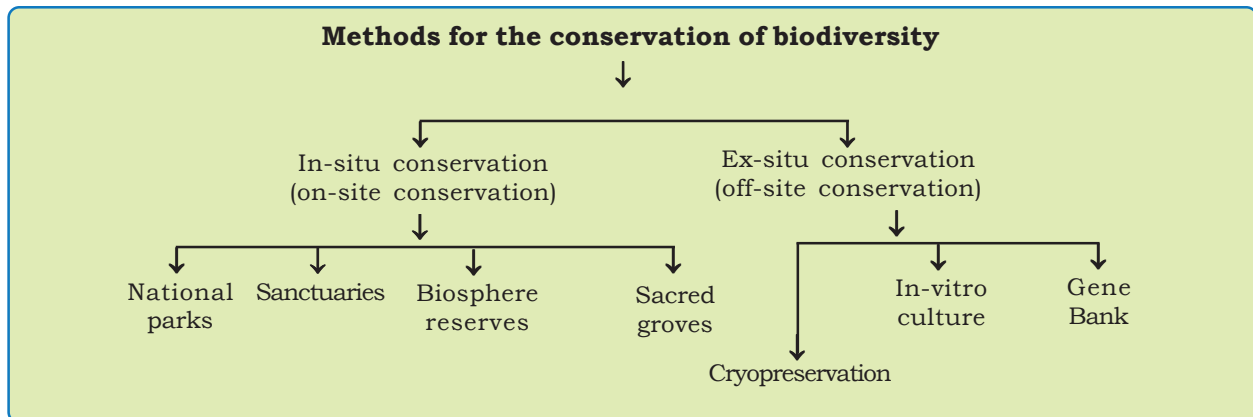
E.g.2: Illegal introduction of exotic **African catfish**, *Clarias gariepinus*, for aquaculture purposes is posing a threat to the indigenous cat fishes.

E.g.3: When exotic and invasive weeds like the ‘carrot grass’ (*Parthenium*), ‘spanish flag’ (*Lantana*), ‘water hyacinth’ (*Eichhornia*) are introduced into our ecosystems they not only damaged the environment but also threatened the very existence of native species.

4. **Co-extinctions:** In an obligate association between a plant and an animal, if a plant becomes extinct, the animal also becomes extinct as seen in a parasite and host association. If the host becomes extinct, parasite meets the same fate. Another association which explains co-extinction is **plant-pollinator mutualism** where extinction of one invariably leads to the extinction of the other.

VI. Methods of Conservation of Biodiversity?

The conservation of biological diversity has become a global concern. There are basically two main types of conservation options-**In-situ** conservation and **Ex-situ** conservation. In-situ is usually seen as the ‘ideal conservation strategy’. Ex-situ conservation can provide a **backup solution** to in-situ conservation projects.



1. In-situ conservation (On-site conservation)

In-situ conservation is the process of protecting an animal species in its natural habitat. The benefit is that it maintains recovering populations in the surrounding where they have developed their distinctive properties. Conservationists identified certain regions by name ‘**Biodiversity hot spots**’ for maximum protection as they are characterized by very high levels of species richness & high degree of endemism. By definition ‘Biodiversity hot spot’ is a ‘Biogeographic Region’ with a significant reservoir of biodiversity that is under threat of extinction from humans. They are Earth’s biologically ‘richest’ and ‘most threatened’ **terrestrial** Ecoregions.

The concept of Biodiversity hot spots was proposed by **Norman Myers**. There are about 34 biodiversity hot spots in the world. As these regions are threatened by destruction, habitat loss is accelerated e.g. :I) Western Ghats and Srilanka ; II) Indo

Burma; III) Himalayas in India. Ecologically unique and biodiversity rich regions are legally protected as in 1. **Biosphere Reserves**–18 (18th is PANNA in Madhya Pradesh), 2. **National Parks**–90, 3. **Sanctuaries**–448.

i. Biosphere Reserves

An area which is set aside, minimally disturbed for the conservation of the resources of the biosphere is '**Biosphere reserve**'. Latest biosphere reserve (17th biosphere reserve in India) is **Seshachalam hills**.

ii. National Parks

A National Park is a natural habitat strictly reserved for protection of natural life. National Parks, across the country, offer a fascinating diversity of terrain, flora and fauna. Some important National Parks in India are - **Jim Corbett National Park** (the first National Park in India located in Uttarakhand), **Kaziranga National Park** (Assam), **Kasu Brahmananda Reddy National Park**, **MahavirHarinaVanasthali National Park** (AP), **Keoladeo Ghana National Park** (Rajasthan), etc.

iii. Sanctuaries

Specific **endangered faunal species** are well protected in wildlife sanctuaries which permits eco-tourism (as long as animal life is undisturbed). Some important Sanctuaries in India (AP) include-Koringa Sanctuary, Eturnagaram Sanctuary, Papikondalu Sanctuary.

iv. Sacred Groves

1. A smaller group of trees than a forest is called grove.
2. A grove of trees of special religious importance to a particular culture is called sacred grove.
3. In these regions all the trees of wild life are venerated (respected) and given total protection.

The following is a list of Sacred Groves in INDIA

Name	State
Khasi and Jaintia Hills	Meghalaya
Aravalli Hills	Rajasthan and Gujarat
Western Ghat region	Karnataka and Maharashtra
Sarguja, Bastar	Chhattisgarh
Chanda	Madhya Pradesh

In Meghalaya, Sacred Groves are the last refuges for a large number of rare and threatened species.

2. Ex-situ conservation:

In ex-situ conservation threatened animals are taken out of their natural habitat and placed in special settings where they are protected. This includes Zoological Parks. Advancement in ex-situ preservation techniques such as **cryopreservation** are helping us protect endangered species (cryopreservation is the preservation of, for example, gametes, embryos of threatened species, etc., at -196°C). In vitro culture, gene banks are mostly used for plants.

VII. IUCN Red data books

International Union for the Conservation of Nature and Natural Resources (IUCN) is the world's main Authority on the issues of conservation status of species.

All the threatened species are listed in the **Red Data Books** published by the IUCN. These species are classified into different categories based on degree of risk and they are chiefly :

- a) Critically endangered
- b) Endangered
- c) Vulnerable

VIII. Conservation of wild life in INDIA

This is through legislation, preservation and organisations

1. Legislation: Under the provision of the **Wildlife Act of 1972**, killing endangered wild animals is strictly prohibited. Trading wildlife products (like tusks, rhino's horns, etc.) is a punishable offence.

2. Preservation: National Parks, Sanctuaries, Biosphere Reserves, Sacred Groves etc. are different regions which are earmarked to protect diverse fauna and flora.

3. Organisational Protection: Organisations which are set up to prevent destruction of India's wild life are

1. Wild life protection society of India (**Dehradun**)
2. Zoological Survey of India (**Kolkata**)

Conservation of biodiversity is a **global necessity**. It is the collective responsibility of all nations to protect the diverse living forms on the planet. One such step was in the form of **EARTH SUMMIT** (1992-Rio de Janeiro) and the other being **WORLD SUMMIT** on sustainable development (Johannesburg-South Africa). They focussed on significant reduction in the current rate of loss of biodiversity at the global, regional and local levels. Efforts must be intensified to pass on our biological legacy to the future generations.

IX. Threatened Species in India

The Asiatic lion	<i>Panthera leo persica</i>
The black buck	<i>Antelope cervicapra</i>
Red panda	<i>Ailurus ochraceus</i>
The lion-tailed macaque	<i>Macaca silenus</i>
Tiger	<i>Panthera tigris</i>
Kashmiri stag	<i>Cervus elaphus hanglu</i>
Elephant	<i>Elephas maximus indicus</i>
Pygmy hog	<i>Sus salvanius</i>
Siberian crane	<i>Grus leucogeranus</i>
Slender loris	<i>Loris tardigradus</i>

In conclusion-Nature is a repository of diverse life. Intrusion into nature's domain distorts its equilibrium. Preserving nature is a collective 'Global Responsibility'. Man should conserve and protect nature in his own interest and for the future generations.

GLOSSARY

Nutrition: The process by which an organism assimilates food and uses it for growth and maintenance.

Protein: Building blocks of life which are polymer chains made of amino acids linked together by peptide bonds.

Prokaryotes: Single celled organisms that lack a membrane bound nucleus.

Eukaryotes: Organisms whose cells contain a membrane bound nucleus and other complex membrane bound cell organelles.

Glycogen: It is a polysaccharide that is the principal storage of glucose in animal cells analogous to starch in plants.

Chordata: A phylum which includes animals showing notochord at some stage in life cycle.

Phylogeny: The evolutionary history of an organism.

Analogous Characters: They denote similarity in function without necessary anatomical similarity (wings of birds and butterflies).

Assortative mating: It is a selective mating/ non-random mating within their population.

Tissue: A group of cells of similar origin that perform a similar function.

Holozoic: It is a method of nutrition that involves ingestion of liquid or solid organic material.

Ctenophora: An invertebrate phylum of animals which are commonly called 'comb jellies'.

Spiral cleavage: It is characteristic of the protostomes in which planes of cell division are diagonal to the vertical axis/polar axis of the embryo.

VERY SHORT ANSWER TYPE QUESTIONS

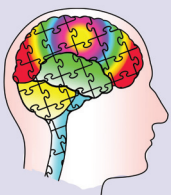
1. Define the term metabolism. Give any one example.
2. How do you differentiate between growth in a living organism and non-living object?
3. What is biogenesis?
4. Define the term histology. What is it otherwise called?
5. Distinguish between embryology and ethology.
6. In a given area, remains of animal that lived in the remote past are excavated for study. Which branch of science is it called?
7. 'Zoos are tools for classification' Explain.
8. Where and how do we preserve skeletons of animals, dry specimens etc?
9. What is trinomial nomenclature? Give an example.
10. What is meant by tautonymy? Give two examples.
11. Differentiate between Protostomia and Deuterostomia.
12. 'Echinoderms are enterocoelomates' Comment.
13. What does ICZN stand for?
14. Give the names of any four protostomian phyla.
15. What is ecological diversity? Mention the different types of ecological diversities.
16. Define species richness.
17. Mention any two products of medicinal importance obtained from Nature.

18. Invasion of an Alien species leads to extinction of native species. Justify this with two examples.
19. List out any four sacred groves in India.
20. Write the full form of IUCN. In which book threatened species are enlisted.

SHORT ANSWER TYPE QUESTIONS

1. Explain the phylogenetic system of biological classification.
2. Explain the hierarchy of classification.
3. What is meant by classification? Explain the need for classification.

4. Define species. Explain the various aspects of 'species'.
5. What is genetic diversity and what are the different types of genetic diversity?
6. What are the reasons for greater biodiversity in the tropics?
7. What is the 'evil quartet'?
8. Explain in brief 'Biodiversity Hot Spots'.
9. Explain '**Rivet Popper**' hypothesis.
10. Write short notes on In-situ conservation.



For ignited minds

- Not For Evaluation

1. If you drink a 'sugar rich' soft drink , the sugar levels in your blood increase quickly. If the sugar levels in blood increase sensitive cells such as 'brain cells' are damaged. Your pancreatic islets produce a hormone called 'insulin' which allows uptake of sugars by body cells so that the body cells utilise them and generate ATP for cellular life activities. The sugar levels of your body fluids come down to normal 'tolerable levels'. What do you think is the 'biological phenomenon' that happened in this case?

Clue: The body always tries to maintain equilibrium state.

2. What, in your expectation, will happen when 'cancer cells' are starved?

Answer: When cancer cells are starved, they cannot show growth .If there is no growth, cells do not multiply. Anabolism is a necessary part of life.

3. What, in your view are reasons for the occurrence of 'variations' in living organisms and the role played by such variations in nature?
4. Why, in your view, children are not 'exact copies' of their parents, when they do inherit parental DNA, which governs their body features?
5. On a cold winter day when you walk outside , you tend to 'shiver'. Can you explain why such 'shivering' is caused?
6. You have learnt about ageing (senescence) in your lesson. Would you like to know one more interesting concept about ageing - It is also controlled by **genes** or to put it in other words **ageing** is also '**genetically**

programmed '. Have you heard of '**Dolly**' the lamb produced by '**cloning**' of the 'udder cell' (somatic cell of the mammary glands) of **Finn Dorset Ewe** (Finn Dorset is a variety of sheep in England ; Ewe is a 'Female sheep'). The average life span of Finn Dorset sheep is about **12** years by which time it develops 'senescence' and dies. 'Dolly' was developed **asexually** by **cloning**, like how a single zygote cell develops into a young one in the case of sexual reproduction. **Do you know how long 'Dolly' lived ?** It is just about **six years**.

Can you guess why it lived for only **six years**, when the average life span of that type of sheep is **twelve** years? *The answer is-*'Dolly', when born, was already **six years old**, '**genetically speaking**', as it developed from the udder cell of a **six year** old sheep, the Finn Dorset ewe. When Dolly was 6 years old, its genetic material is 12 years old, hence the death .

7. Do you agree with the statement the log-log scale linear relation in the south American tropical forests will be $> 45^\circ$?
8. When the human zygote undergoes the first division into two cells, if the two cells are separated they develop into identical twins. Will the same thing happen in the case of earthworm? Why/ Why not?
9. In your opinion can a population be made up of a group of intra-specific variants'? if your answer is Yes, how do you defend?
10. Which among the following encompasses the others - Genus / Grade / Division / Class / Order?